Supplementary Information

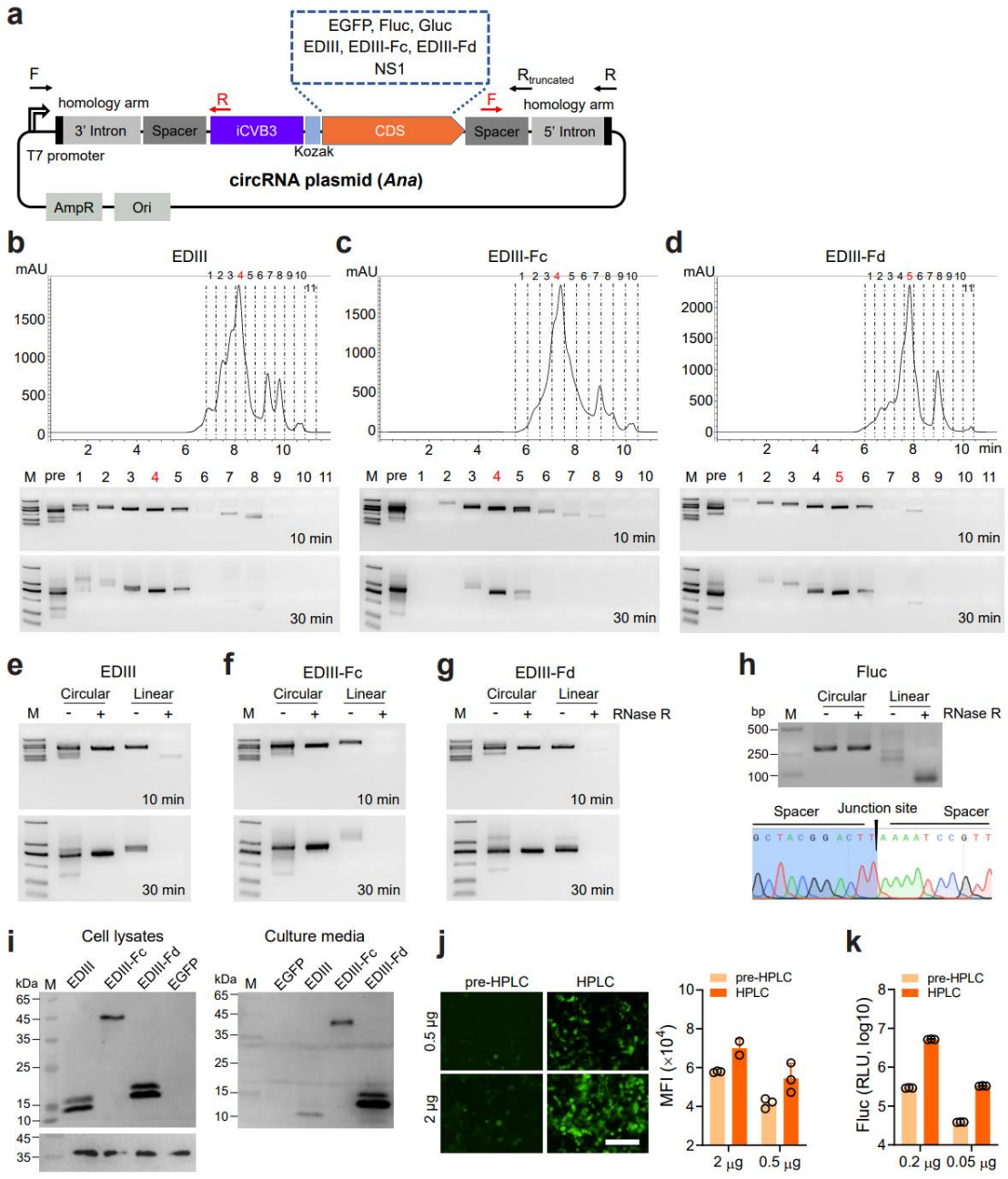
A single-dose circular RNA vaccine prevents Zika virus infection without enhancing dengue severity in mice

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Supplementary Methods

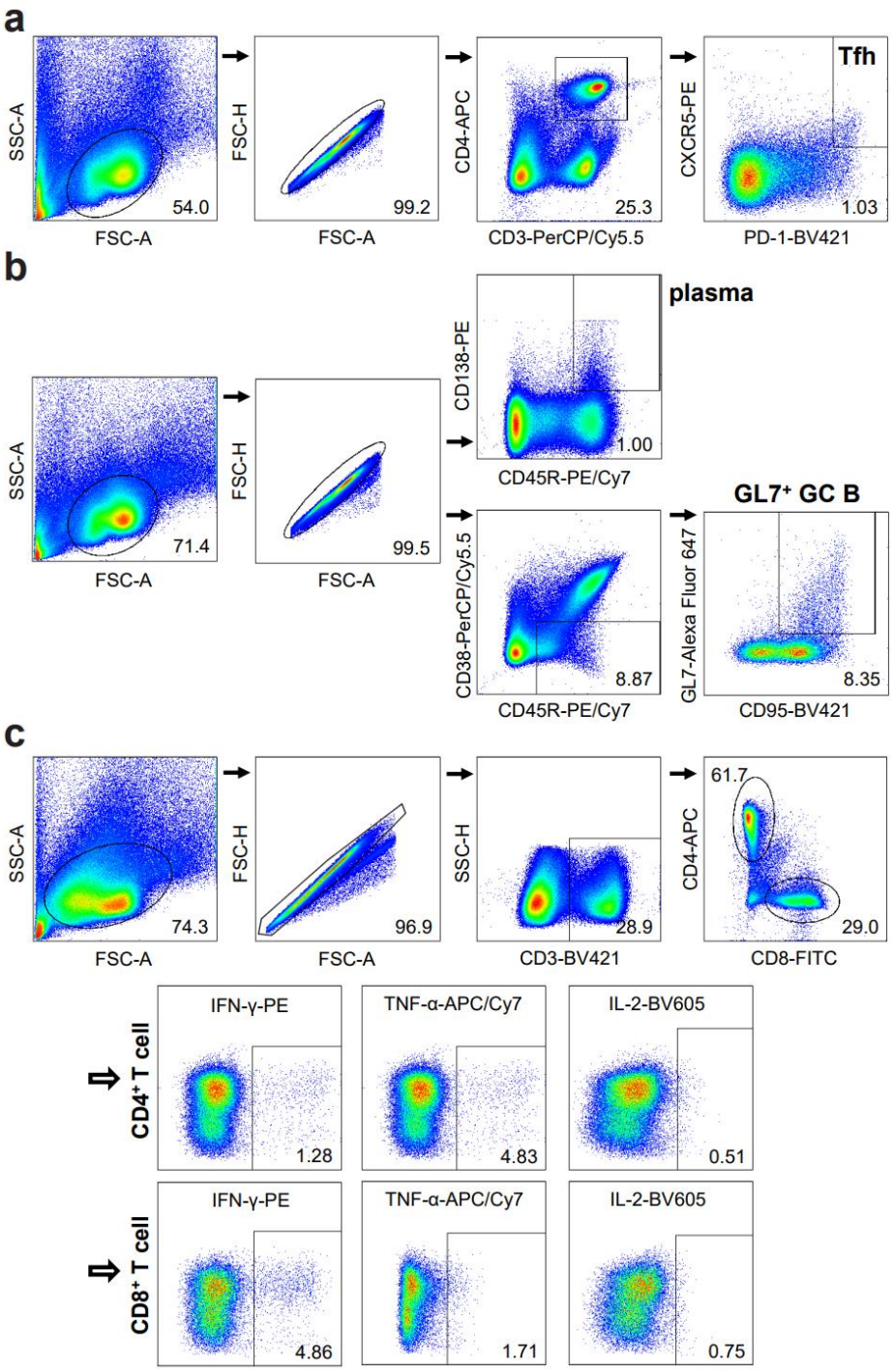
Production of modified linear mRNAs. mRNAs encoding EDIII-Fc and NS1 were synthesized by *in vitro* transcription. In brief, plasmid templates covering the tPA signal sequence and the coding sequences for EDIII-Fc and NS1 were constructed, linearized, and transcribed *in vitro* by T7 RNA polymerase. During the transcription process, co-transcriptional capping was employed to add cap analogs to the 5' end of the mRNA in real time. Meanwhile, modified nucleobase N1-Methylpseudouridine-5-Triphosphate (m1ψ) was incorporated in place of the standard uridine-5-Triphosphate. Following *in vitro* transcription, mRNA products were purified through the lithium chloride (LiCl) method and dissolved in RNase-free water. Finally, purified mRNAs were analyzed by agarose gel electrophoresis, quantified by spectrophotometry, and subjected to encapsulation by LNPs.

Supplementary Figures

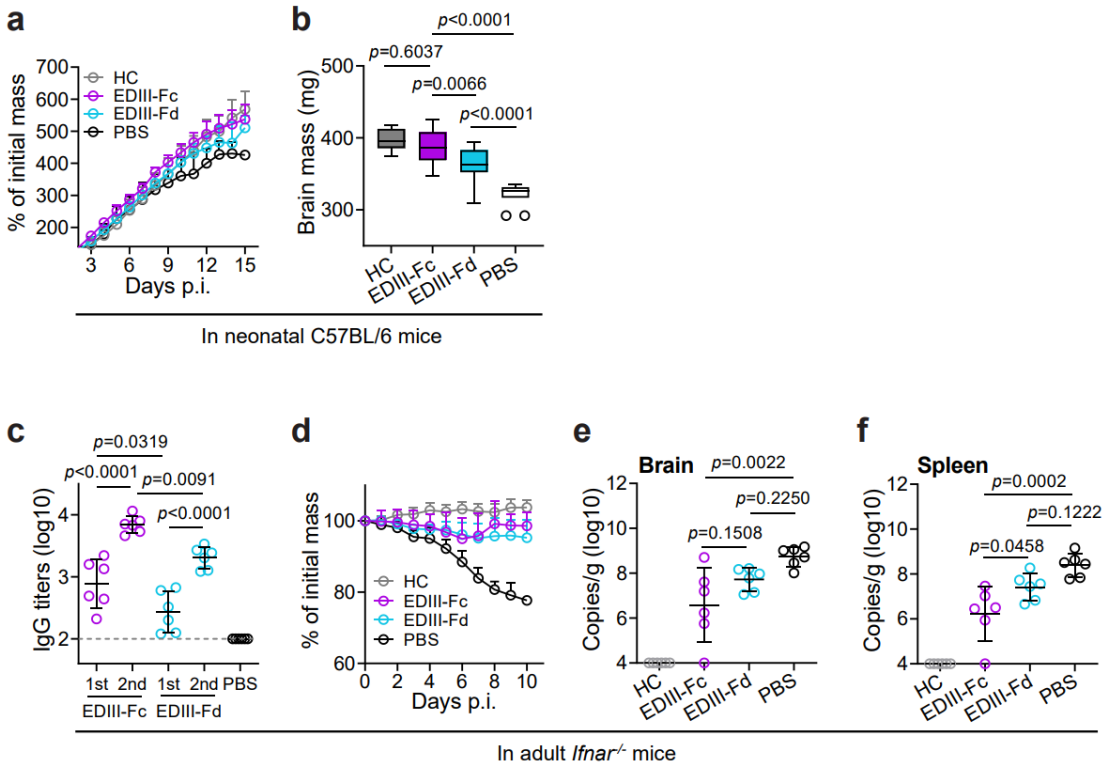


Supplementary Fig. 1 (Related to Fig. 1). Generation and characterization of circRNAs based on Ana PIE method. a, Schematic diagram of the circRNA plasmids (Ana). Fragments containing each indicated element were inserted downstream of the T7 promoter in pGEM-T-easy vector. Coding sequences for EDIII, EDIII-Fc, EDIII-Fd, and NS1 were fused with the signal sequence of human tissue plasminogen activator (tPA) at 5' termini. Black arrows, primers for amplifying the DNA templates for circRNA precursors (F, R) and control linear precursors (F, $R_{truncated}$). Red arrows, primers for amplifying junction site. **b-d**, Purification of the circRNAs encoding EDIII

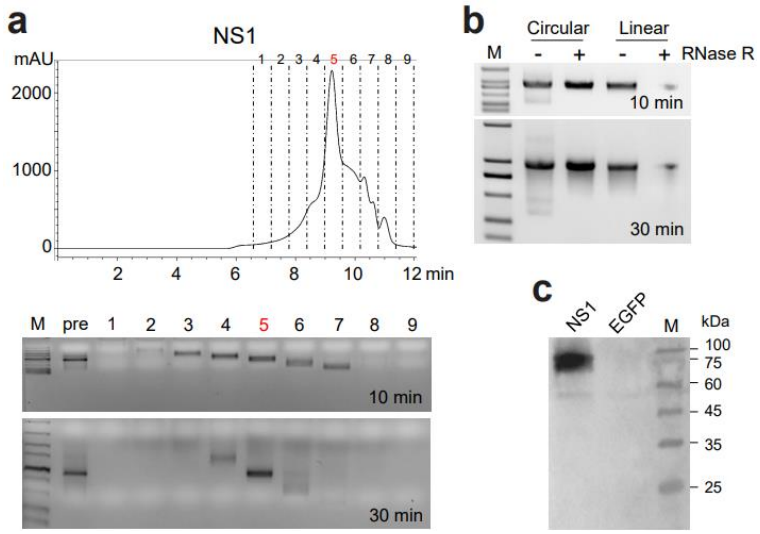
(**b**), EDIII-Fc (**c**), and EDIII-Fd (**d**). Graphs of HPLC and images of agarose-gel electrophoresis captured at 10 or 30 min are shown. Collected fractions have been indicated. Peak fractions containing the majority of circRNAs are marked in red. M, DNA marker DL2000, from top to bottom: 2000, 1000, 750, 500, 250, 100 bp. **e**-**g**, RNase R digestion of the circRNAs encoding EDIII (**e**), EDIII-Fc (**f**), EDIII-Fd (**g**), and their respective linear precursors. Images of agarose-gel electrophoresis captured at 10 and 30 min are shown. **h**, Junction site PCR and Sanger sequencing. circRNAs and control linear precursors were treated with or without RNase R digestion. PCR products were analyzed by agarose-gel electrophoresis and Sanger sequencing. Target fragment is 223 bp. Shown are the results for Fluc circRNA. Results for other circRNAs are similar. **i**, Western blot analysis of EDIII, EDIII-Fc, and EDIII-Fd in cell lysates and culture media of the transfected HEK293T cells using EDIII protein immune mice sera in the presence of β-mercaptoethanol. Molecular weights of monomeric EDIII, EDIII-Fc, and EDIII-Fd are 12, 38, and 15 kDa, respectively. GAPDH in cell lysates was also analyzed. **j**, **k**, Expression of EGFP in A549 cells (**j**) and Fluc in HEK293T cells (**k**) that were transfected with EGFP or Fluc circRNA before or after HPLC purification. Scale bar = 200 μm. Data points in (**j**, **k**) represent three technical replicates and presented as mean ± s.d. Data are representative of at least two independent experiments with similar results. Source data are provided as a Source Data file.



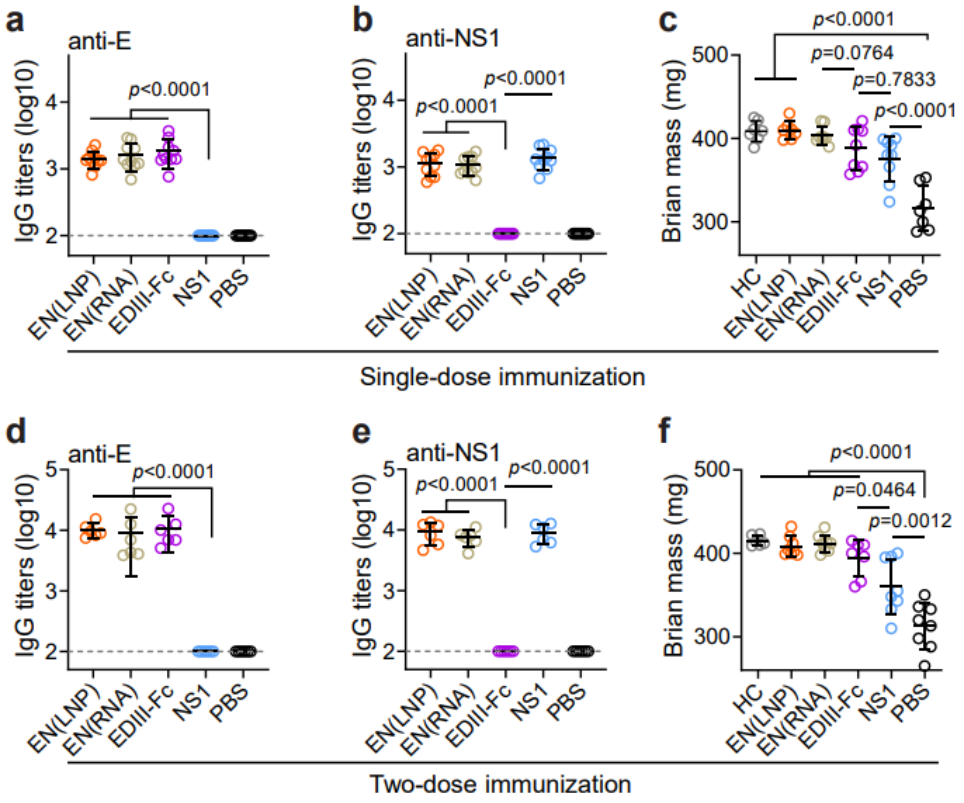
Supplementary Fig. 2 (Related to Fig. 1). Gating strategies for lymphocyte analysis. **a**, Gating strategy for Tfh cells in the ILNs. Tfh cell designation criteria: $CD3^+$, $CD4^+$, $CXCR5^+$, $PD\text{-}1^+$. **b**, Gating strategy for plasma cells and $GL7^+$ GC B cells in the ILNs. Plasma cell designation criteria: $CD138^+$, $CD45R^+$. $GL7^+$ GC B cell designation criteria: $CD38^-$, $CD45R^+$, $CD95^+$, $GL7^+$. **c**, Gating strategy for $CD4^+$ T and $CD8^+$ T cells that secret IFN-γ, TNFα, or IL-2 in the spleens.



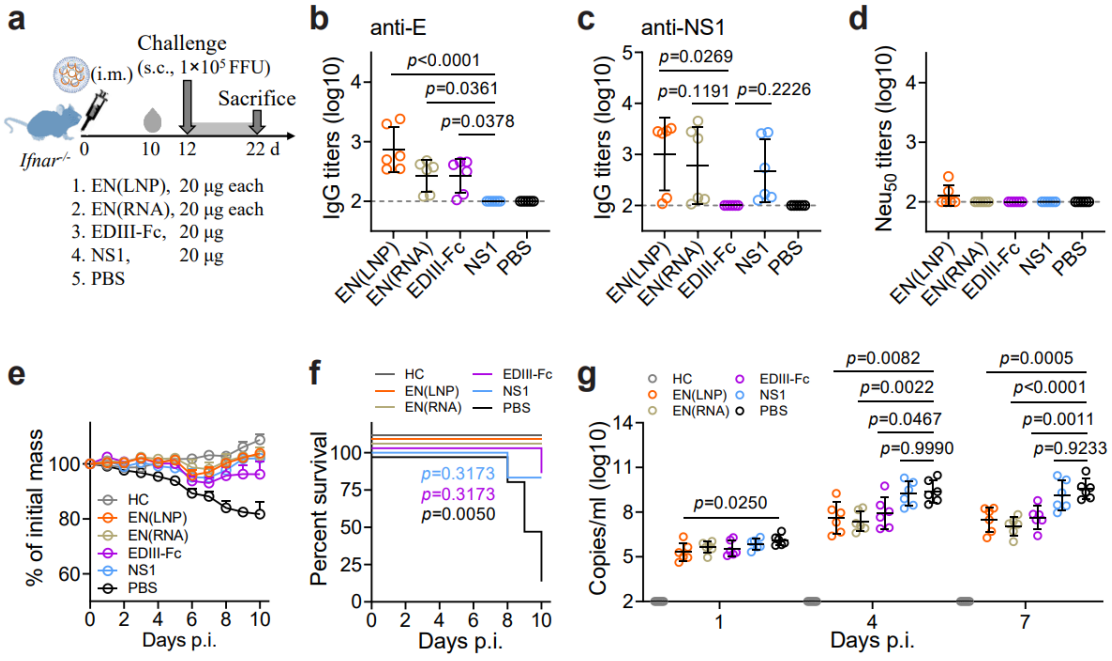
Supplementary Fig. 3 (Related to Fig. 2). Immunogenicity and protective effects of EDIII-Fc and EDIII-Fd circRNAs. a, Growth curves of the pups after ZIKV challenge. **b**, Brain masses of the pups at sacrifice. Box plots indicate median (middle line), 25-75 percentile (box), 5-95 percentile (whiskers) and outliers (single points). **c**, Anti-E IgG titers in *Ifnar*$^{-/-}$ mice at 2 weeks after each immunization. Serially diluted mice sera were measured by ELISA. IgG titers were calculated as the reciprocals of the highest dilutions at which the optical density values at 450 nm (OD450) were equal to or higher than cut-off values. Limit of detection (LOD) is 100 and marked by gray dashed lines. **d**, Body masses of the *Ifnar*$^{-/-}$ mice. **e**, **f**, ZIKV genome copies in the brains (**e**) and spleens (**f**) of the *Ifnar*$^{-/-}$ mice at sacrifice. Data points represent mean values for one group (**a**, **d**), or mean values of two (**c**) or three (**e**, **f**) technical replicates for one mouse. n = 13 (Healthy control, HC), 14 (EDIII-Fc), 16 (EDIII-Fd) and 10 (PBS) for (**a**, **b**). n = 6 for (**c**-**f**). Data are representative of two independent experiments and presented as mean ± s.d. Comparisons were performed by one-way ANOVA and Tukey's multiple comparison tests. *p* values are shown on the graphs. Source data are provided as a Source Data file.



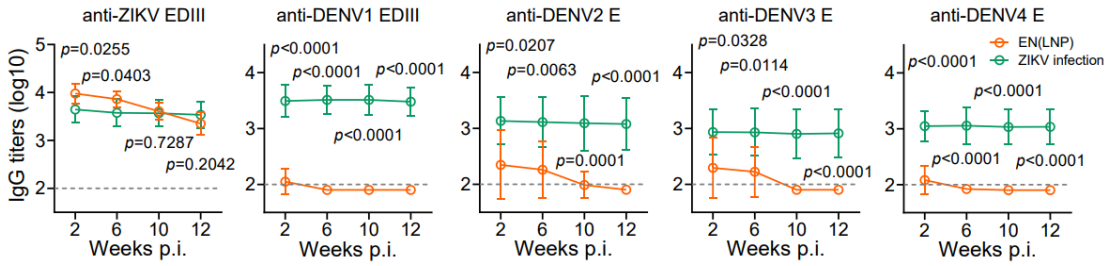
Supplementary Fig. 4 (Related to Fig. 3 and Fig. 4). Generation of ZIKV NS1 circRNA. a, Purification of NS1 circRNA. Graph of HPLC and images of agarose-gel electrophoresis captured at 10 and 30 min are shown. Collected fractions have been indicated. Peak fraction containing the majority of NS1 circRNA is marked in red. M, DNA marker DL5000, from top to bottom: 5000, 3000, 2000, 1500, 1000, 750, 500, 250, 100 bp. **b**, RNase R digestion of NS1 circRNA. Images of agarose-gel electrophoresis captured at 10 and 30 min are shown. Here M is DNA marker DL2000. **c**, Western blot analysis of the culture media of HEK293T cells transfected with NS1 circRNA using anti-ZIKV NS1 mAb 749-A4 in the absence of β-mercaptoethanol. Molecular weight of dimeric NS1 is ~ 80 kDa. Data are representative of at least two independent experiments with similar results. Source data are provided as a Source Data file.



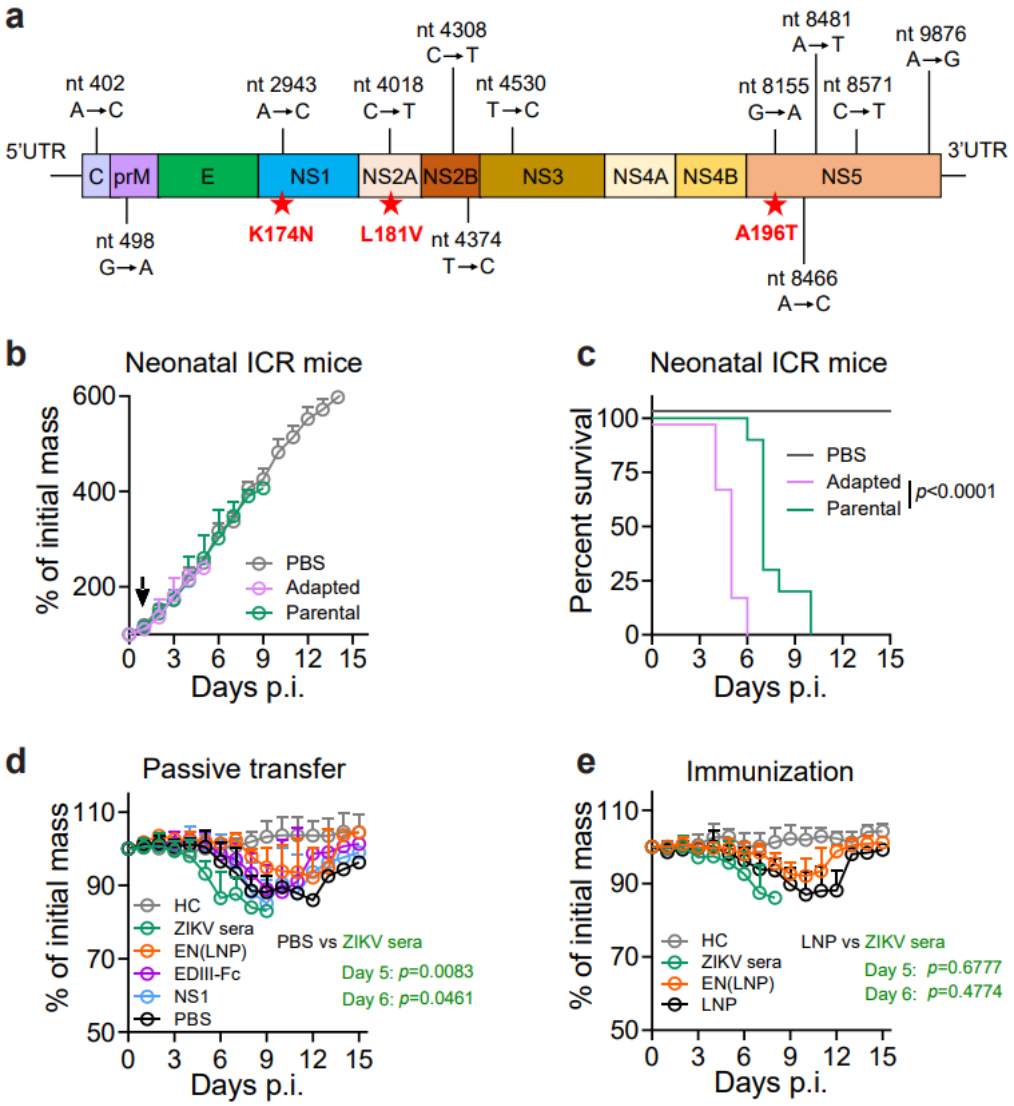
Supplementary Fig. 5 (Related to Fig. 3 and Fig. 4). Immunogenicity and protective efficacy of the combination of circRNAs encoding EDIII-Fc and NS1. a, **b**, Anti-E (**a**) and anti-NS1 (**b**) IgG titers at 2 weeks after the first immunization. IgG titers in immune mice sera (as shown in Fig. 3a, n = 10 per group) were measured by ELISA. LODs are 100 and marked by gray dashed lines. **c**, Brain masses of the pups at sacrifice. n = 7 (HC, EN(LNP)), 8 (EN(RNA) , PBS) and 9 (EDIII-Fc, NS1). **d**, **e**, Anti-E (**d**) and anti-NS1 (**e**) IgG titers at 2 weeks after the second immunization. n = 6 per group. **f**, Brain masses of the pups at sacrifice. n = 6 (HC), 7 (EDIII-Fc) and 8 (EN(LNP), EN(RNA), NS1, PBS). Data points represent mean values of two technical replicates for one mouse (**a**, **b**, **d**, **e**), or values for individual mice (**c**, **f**). Data are representative of two independent experiments and presented as mean ± s.d. Comparisons among groups were performed by one-way ANOVA and Tukey's multiple comparison tests. *p* values are shown on the graphs. Source data are provided as a Source Data file.



Supplementary Fig. 6 (Related to Fig. 3 and Fig. 4). Immunogenicity and protective efficacy of a single-dose circRNA vaccination in adult *Ifnar*$^{-/-}$ mice. a, Schematic diagram of immunization and challenge assay. Twelve-week-old mice (n = 6 per group, male and female) were i.m. immunized once with EN(LNP), EN(RNA), EDIII-Fc circRNA, or NS1 circRNA at the indicated doses. Ten days later, mice sera were collected. At 12 days after immunization, mice were s.c. challenged with 1×10^5 FFU of ZIKV. **b-d**, Anti-E IgG (**b**), anti-NS1 IgG (**c**), and nAb (**d**) titers. IgG and nAbs were measured by ELISA and FNT, respectively. LODs are 100 and marked by gray dashed lines. **e**, Body masses of the mice after challenge. **f**, Survival curves of the mice after challenge. Mice that lost 20% of initial body masses were considered dead and euthanized. Comparisons were conducted between each group and HC. **g**, ZIKV genome copies in the sera at 1, 4, and 7 days after challenge. Data points represent mean values of two (**b-d**) or three (**g**) technical replicates for one mouse, or mean values for one group (**e**). Data are representative of two independent experiments and presented as mean ± s.d. Comparisons were performed by one-way ANOVA and Tukey's multiple comparison tests (**b**, **c**, **d**, **g**) or Log-rank (Mantel-Cox) tests (**f**). *p* values are shown on the graphs. Source data are provided as a Source Data file.

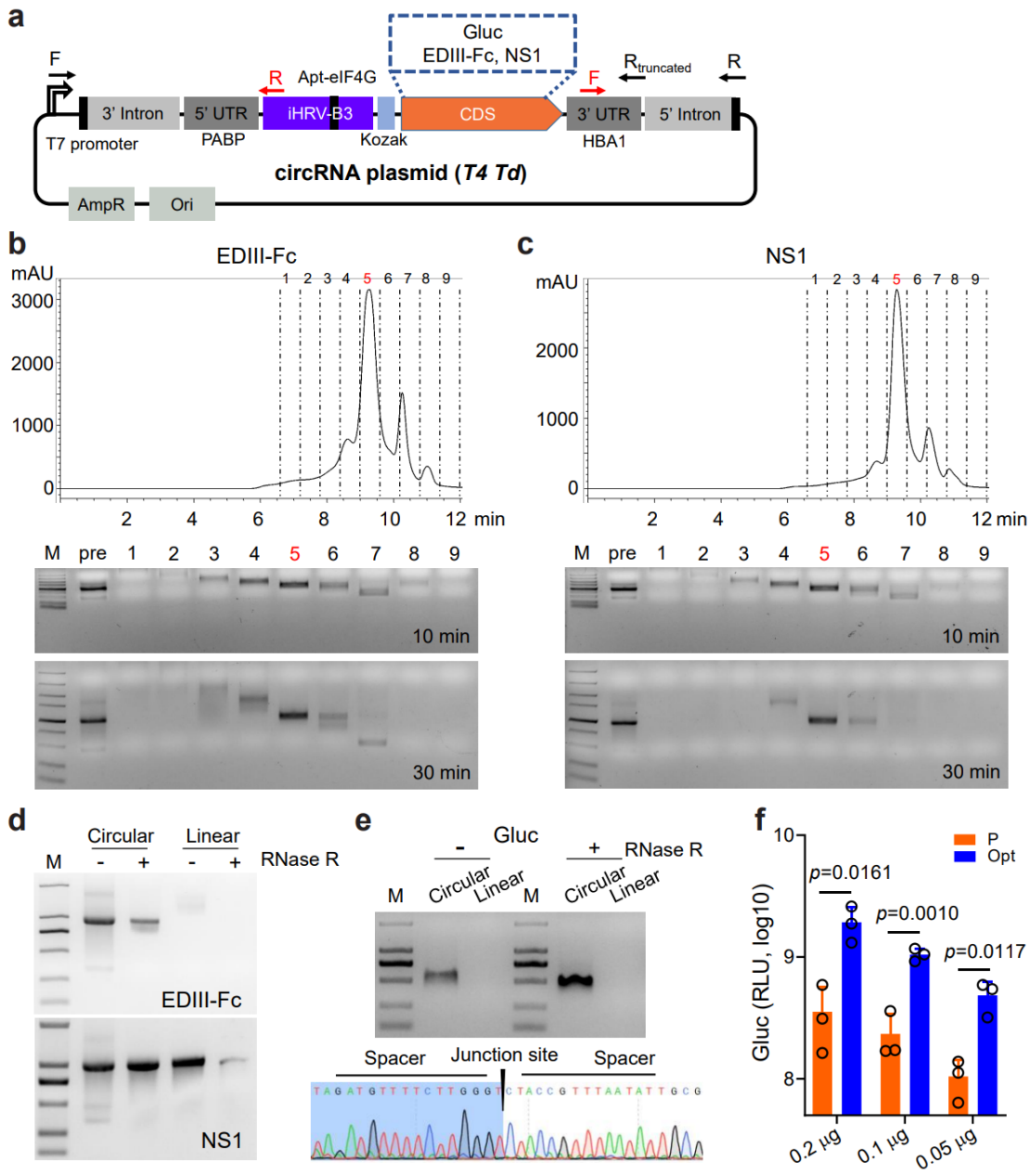


Supplementary Fig. 7 (Related to Fig. 5). Kinetics of cross-reactive antibodies elicited by EN(LNP) immunization or ZIKV infection. Eight-week-old female C57BL/6 mice were i.m. immunized with EN(LNP) (20 μg each circRNA per mouse, n = 8) or s.c. infected with ZIKV (1×10^5 FFU per mouse, n = 6). At 2, 6, 10, and 12 weeks after immunization or infection, mice sera were collected. Titers of IgG antibodies against ZIKV and DENV were examined by ELISA. LODs are 100 and marked by gray dashed lines. Data are representative of at least two independent experiments and presented as mean ± s.d. Comparisons were performed by unpaired two-tailed Student's t-test. *p* values are shown on the graphs. Source data are provided as a Source Data file.



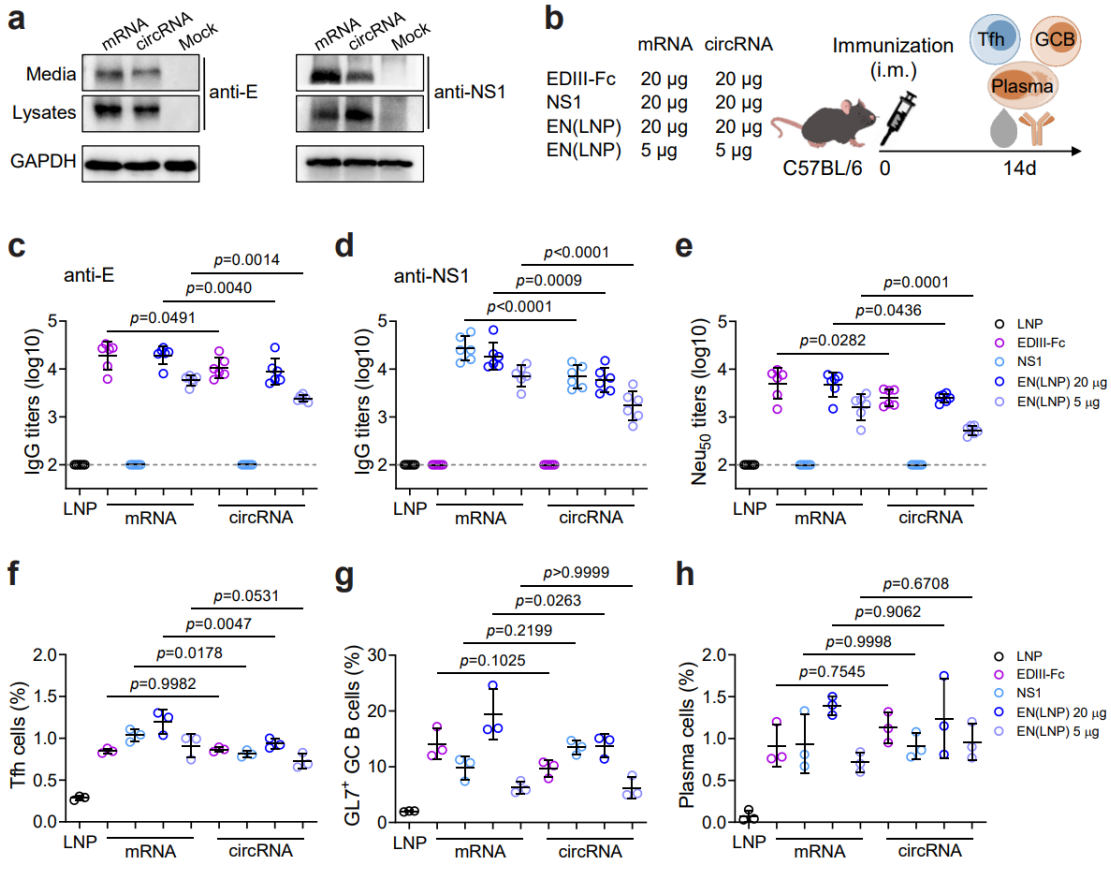
Supplementary Fig. 8. In vivo assessment of the ADE risks of circRNA vaccines on DENV2 infection (Related to Fig. 5). a, Mutations in the genome of mouse-adaptive DENV2 in comparison with the reference strain. DENV2 strain 16681 was alternatively passaged between 1-day-old C57BL/6 pups and Vero cells for 3 cycles. Viral genomes were extracted and subjected to next-generation sequencing and Sanger sequencing. Substitutions of nucleotides compared to the genome sequence of original reference strain are indicated. Non-synonymous mutations are marked by red asterisks and the locations in viral proteins are indicated. **b**, Growth curves of 1-day-old ICR mice that were infected with 1×10^4 FFU of parental or mouse-adapted DENV2 through an intracerebral route. Black arrow indicates the time point of challenge. **c**, Survival curves 1-day-old ICR mice after DENV2 challenge. n = 10 per group for (**b**, **c**). **d**, Body masses of DENV2-infected *Ifnar*$^{-/-}$ mice that were passively transferred with immune sera (n = 5 per group, male and female, as shown in Fig. 5c) prior to challenge. **e**, Body masses of DENV2-infected *Ifnar*$^{-/-}$ mice that received EN(LNP) immunization (n = 6 per group,

male and female, as shown in Fig. 5f) prior to challenge. Data are representative of two independent experiments and presented as mean ± s.d. Comparisons were performed by Log-rank (Mantel-Cox) tests (**c**), or by one-way ANOVA and Tukey's multiple comparison tests (**d**, **e**). *p* values are shown on the graphs. The *p* values in (**d**, **e**) indicate the difference between ZIKV sera group and either PBS (**d**) or LNP (**e**) group at the indicated time points. *p* values are shown on the graphs. Source data are provided as a Source Data file.

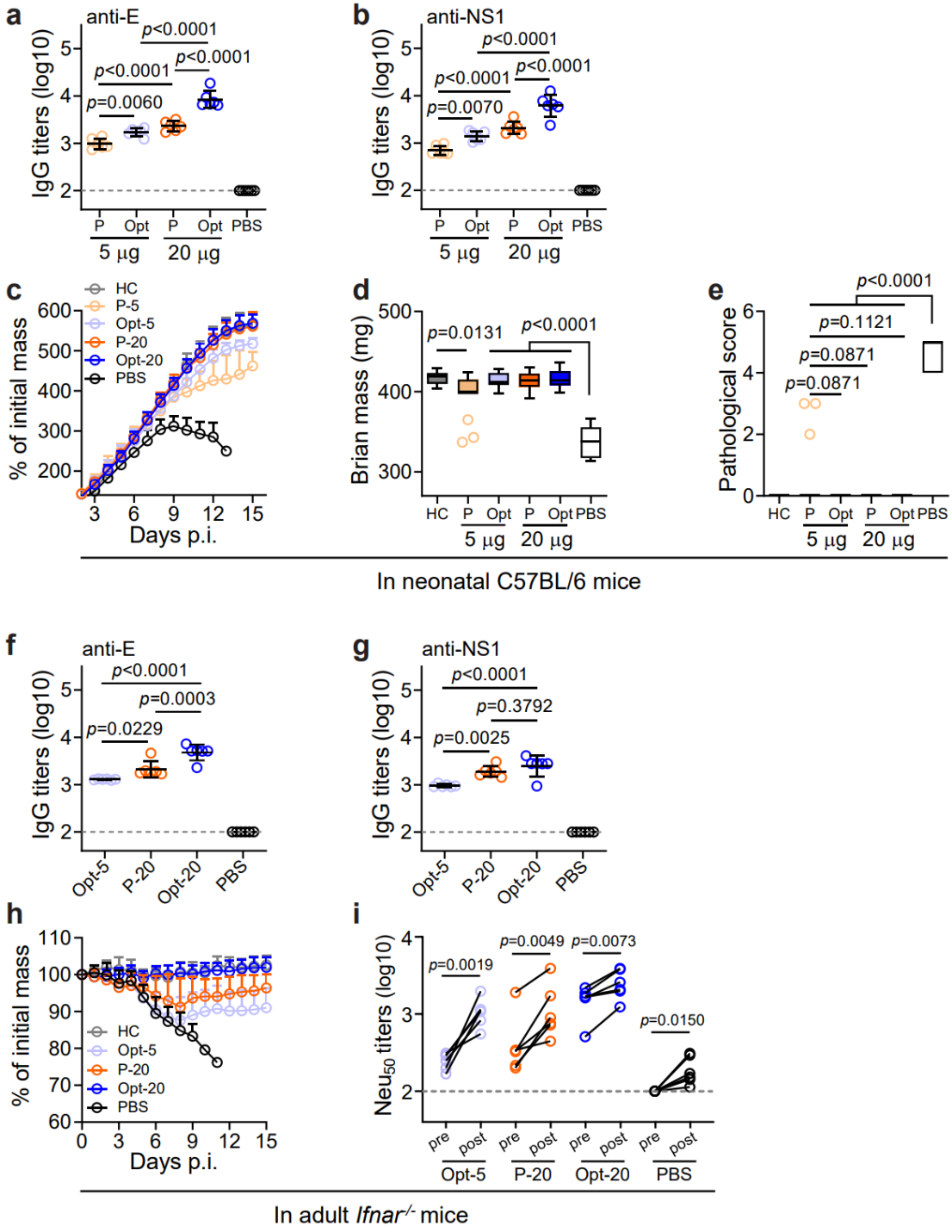


Supplementary Fig. 9 (Related to Fig. 6). Generation of optimized circRNAs. a, Schematic diagram of the optimized circRNA plasmid (T4 Td). Each element has been indicated. The coding sequences for EDIII-Fc and NS1 were fused with tPA signal sequence at 5' termini. Black arrows, primers for amplifying the DNA templates for circRNA precursors (F and R) and control linear precursors (F and $R_{truncated}$). Red arrows, primers for amplifying junction site. **b**, **c**, Purification of optimized circRNAs encoding EDIII-Fc (**b**) and NS1 (**c**). Graphs of HPLC and images of agarose-gel electrophoresis captured at 10 or 30 min are shown. Collected fractions are indicated. Peak fractions containing the majority of circRNAs are marked in red. M, DNA marker DL5000. **d**, RNase R digestion of the optimized circRNAs encoding EDIII-Fc or NS1, or their respective linear precursors. Images of agarose-gel electrophoresis captured at 30 min

are shown. M, DNA marker DL2000. **e**, Junction site PCR and Sanger sequencing. Optimized circRNAs and control linear precursors were treated with or without RNase R digestion and subjected to junction site PCR. Products were analyzed by agarose-gel electrophoresis and Sanger sequencing. Target fragment is 303 bp. Shown are the results of optimized Gluc circRNA. Results of other optimized circRNAs are similar. M, DNA marker DL2000. **f**, Gluc activities in the culture media. HEK293T cells in 96-well plates were transfected with prototype (P) or optimized (Opt) Gluc circRNA at 0.2, 0.1, or 0.05 μg per well. One day later, Gluc activities in the culture media were measured. Data points in (**f**) represent three independent repeats and presented as mean ± s.d. Data are representative of at least two independent experiments with similar results. *p* values are shown on the graphs. Source data are provided as a Source Data file.



Supplementary Fig. 10 (Related to Fig. 6). Comparative analysis of the antigen expression and immunogenicity of m1ψ modified mRNA and circRNA. a, Western blot analysis of EDIII-Fc and NS1 in cell lysates or culture media of the transfected HEK293T cells. Analysis was performed using EDIII protein immune mice sera or ZIKV sera in the absence of β-mercaptoethanol. GAPDH was analyzed in parallel. This experiment was performed twice with similar results. **b**, Schematic diagram of immunization assay. The vaccines, doses, and experimental schedule are indicated. **c-e**, Anti-E IgG (**c**), anti-NS1 IgG (**d**) and anti-ZIKV nAb titers (**e**) at 14 days after immunization. Serially diluted mice sera (n = 6 per group) were measured by ELISA and FNT. LODs are 100 and marked by gray dashed lines. **f-h**, Frequencies of Tfh cells in $CD4^+$ T cells (**f**), activated $GL7^+$ GC B cells in $CD38^-$ B cells (**g**) and plasma cells in lymphocytes (**h**) in the ILNs. At 14 days after immunization, ILN lymphocytes were isolated, labeled with surface marker antibodies, and measured by flow cytometry. n = 3 per group. Gating strategies have been shown in Supplementary Fig. 2. Data points represent mean values of two technical replicates for one mouse (**c-e**) or values for individual mice (**f-h**). Data are representative of two independent experiments and presented as mean ± s.d. Comparisons are performed by one-way analysis of variation (ANOVA) and Tukey's multiple comparison tests. *p* values are shown on the graphs. Source data are provided as a Source Data file.



Supplementary Fig. 11 (Related to Fig. 6). Immunogenicity and protective efficacy of a single-dose optimized circRNA vaccine. a, **b**, Anti-E IgG (**a**) and anti-NS1 IgG (**b**) titers in C57BL/6 mice that received prototype (P) or optimized (Opt) EN(LNP) (n = 6 per group, as shown in Fig. 6c). IgG titers were measured by ELISA. LODs are 100 and marked by gray dashed lines. **c**, Growth curves of the pups after ZIKV challenge. **d**, Brain masses of the pups at sacrifice. **e**, Pathological scores of neonatal brains. Meningeal lymphocyte infiltration and cortex laminar necrosis were scored in a single-blind manner. Box plots in (**d**, **e**) indicate median (middle line), 25-75 percentile (box),

5-95 percentile (whiskers) and outliers (single points). n = 10 (HC), 15 (P-5), 12 (P-20) and 14 (Opt-5, Opt-20, PBS) for (**c**-**e**). **f**, **g**, Anti-E IgG (**f**) and anti-NS1 IgG (**g**) titers in *Ifnar*$^{-/-}$ mice that received prototype or optimized EN(LNP). LODs are 100 and marked by gray dashed lines. **h**, Body masses of the *Ifnar*$^{-/-}$ mice. **i**, anti-ZIKV nAb titers 1 week before or after challenge. LOD is 100 and marked by gray dashed line. n = 6 for **(f-i)**. Data points represent mean values of two technical replicates for one mouse (**a**, **b**, **f**, **g**, **i**), or mean values for one group (**c**, **h**). Data are representative of two independent experiments and presented as mean ± s.d. Comparisons were performed by one-way ANOVA and Tukey's multiple comparison tests (**a**, **b**, **d**, **e**, **f**, **g**) or by paired, two-tailed Student's test (**i**). *p* values are shown on the graphs. Source data are provided as a Source Data file.

Supplementary Table 1. Characterization of LNP formulations used in this study.

Samples		Size (nm)	PDI[a]	EE[b] (%)
empty LNPs		79.6±0.62	0.177±0.009	NA[c]
circRNA	EDIII	83.9±0.28	0.199±0.003	91.16
	EDIII-Fc	87.3±1.09	0.210±0.014	90.38
	EDIII-Fd	83.7±0.96	0.199±0.004	92.05
	NS1	70.4±0.16	0.143±0.004	95.49
	EN(RNA)	89.3±0.07	0.152±0.008	93.78
	Opt-EDIII-Fc[d]	81.3±0.40	0.151±0.008	91.17
	Opt-NS1[d]	85.3±0.30	0.182±0.015	95.72
mRNA	EDIII-Fc	91.0±0.20	0.180±0.010	94.33
	NS1	75.0±0.86	0.146±0.010	93.82

[a]Polydispersity index (PDI)

[b]Encapsulation efficiency (EE)

[c]Not Applicable (NA)

[d]The optimized EN(LNP) encompassed two components: optimized EDIII-Fc circRNA and optimized NS1 circRNA that were encapsulated into LNPs separately.